Lunar tide on a coastal–inland optical-clock link

[Author list to be added.]

[1] Hefei National Laboratory, University of Science and Technology of China, Hefei, China

[2] Shanghai Research Center for Quantum Sciences and CAS Center for Excellence in Quantum Information and Quantum Physics, University of Science and Technology of China, Shanghai, China

[3] Hefei National Research Center for Physical Sciences at the Microscale and School of Physical Sciences, University of Science and Technology of China, Hefei, China

[4] Innovation Academy for Precision Measurement Science and Technology, Chinese Academy of Sciences, Wuhan, China

[5] Jinan Institute of Quantum Technology and CAS Center for Excellence in Quantum Information and Quantum Physics, University of Science and Technology of China, Jinan, China

[6] Time and Frequency Geodesy Center (TFGC), School of Geodesy and Geomatics, Wuhan University, Wuhan, China

Optical clocks measure gravitational potential through their relative rates. A coastal–inland comparison provides a test of time-varying potential in which ocean loading is a substantial part of the prediction. We compare strontium and ytterbium clocks in Shanghai and Wuhan over a 1,350-km fibre link during six interrupted operating periods in 2026. Ocean loading is 34% of the predicted tidal root-mean-square. After removing only period constants, 2,905 frequency averages show a lunar semidiurnal oscillation. The M2 response is 1.00 ± 0.12, with a quadrature coefficient of -0.07 ± 0.18 in a joint phase fit. N2 and a separate loading scale remain unresolved. Coherent solar-day disturbances couple weakly to the M2 estimator. Allowing 24-h and 12-h terms to reset between periods changes the scale to 0.82 ± 0.20; a joint amplitude–phase test then no longer excludes zero. The measurement tests the dominant lunar response on a baseline with substantial predicted ocean loading.

Static chronometric levelling with optical clocks is established.[1,2,3,4,5] A further question is the time-varying potential on a baseline where ocean loading is not a small correction. Shanghai lies on the East China Sea and Wuhan lies inland; for this pair, loading is 34% of the predicted tidal root-mean-square, and the differential is dominated by the lunar semidiurnal tide.[6,7,8,9] Six interrupted operating periods in 2026 test that fixed prediction.

Transportable clocks and coherent optical networks extend gravitational measurements beyond one laboratory.[10,11,12] Astronomical forcing, deformation of the Earth and redistribution of ocean water produce time-variable clock rates.[13,14,15] The coastal–inland geometry gives the solid-Earth and ocean-loading contributions distinct amplitudes and phases. The lunar tide also advances through the solar day, supplying information that is absent from one short observation window.

A coastal–inland test of the lunar prediction

We compare a Sr clock in Shanghai with a Yb clock in Wuhan over a 1,350-km fibre route (Fig. 1a). The related frequency-ratio experiment and evaluations of the clock systems provide the apparatus context.[16,17,18] Long-link coherent dissemination supplies the technical basis; performance reported for other links is not assigned to this record.[19,20] Between 7 January and 26 August 2026, six operating periods provide 2,905 non-overlapping 600-s averages from 1,737,073 nominal one-second members. These span 231 calendar days and accumulate 20.1 nominal sample days in 49 effective intervals (Extended Data Fig. 1 and Table 1). Predictions are averaged over the same retained members as each observation.

The deterministic prediction combines an IERS-based solid-Earth calculation with FES2022 ocean loading.[6,7,8,9] On the retained sampling, their root-mean-square contributions are 3.245×10^{-18} and 1.507×10^{-18}, and the combined value is 4.410×10^{-18}. The loading-to-total RMS ratio is 34.2%, a model prediction rather than a measured loading fraction. In a common astronomical M2 basis, the

solid-Earth and loading vectors differ in phase by 19.9° and combine to an amplitude of 4.78×10^{-18} before the nodal factor (Fig. 1b). This geographic sensitivity motivates a test of the combined tidal potential. The six periods sample different M2 phases at overlapping local clock times (Fig. 1c). This phase diversity supplies leverage against a solar-day disturbance with a common phase; the consequences of phase changes between periods are tested below.

Observed M2 amplitude and phase

After subtracting only six operating-period constants, the observations folded by their predicted M2 phase show a semidiurnal oscillation (Fig. 2a). Twelve fixed 30-degree phase groups contain all 2,905 averages, with 222–266 observations per group. Their maximum is near the predicted M2 maximum. Other tidal components remain in this display; the complete scatter is shown in Extended Data Fig. 2.

To estimate the constituent response, we fit M2 and N2 separately, together with near-12-h and diurnal groups and the six period constants. The ungrouped theoretical remainder is included once. All coefficients are fitted jointly to unchanged observations: individual segment centres are not adjusted to a fitted tide. The resulting M2 scale is

$$A_{M2} = 0.997 \pm 0.123.$$

Errors summarize residual-scaled least-squares, delete-day and delete-period variation as specified in Methods. On summer P6–P8 alone, the identical estimator and error construction give 0.882 ± 0.188 from 1,642 bins and 983,049 nominal members. The earlier three periods give 1.063 ± 0.487. The early-to-summer difference is smaller for M2 than for the whole-template point estimates, 1.087 and 0.704 (Extended Data Table 3); our physical claim therefore concerns the dominant M2 response. The summer subset overlaps the complete record and is not a separate replication.

A joint amplitude–phase fit gives M2 in-phase and quadrature coefficients $(0.998, -0.068)$, with marginal errors $(0.123, 0.177)$ (Fig. 2b). The expected unit, in-phase point is consistent with the observations. Neither timestamps nor an arbitrary template phase are adjusted to obtain this agreement. The folded display and phase fit use the same data and provide complementary views of the measurement.

N2 is not separately resolved: its fixed-phase scale is 0.541 ± 0.648. Constraining M2 and N2 to their predicted ratio gives a common lunar scale 0.973 ± 0.121, while replacing period levels by individual-segment levels gives a separate M2 scale 0.942 ± 0.135 (Extended Data Table 4). Joint estimation and projected removal of the stronger component agree when its fitted uncertainty is propagated. These checks retain the weaker constituent without treating subtraction as a new independent observation.

Solar-day discrimination and measured controls

The distinction between coherent and resetting daily disturbances matters on interrupted observations. A prescribed 12-h sinusoid of fractional-frequency amplitude 10^{-18}, with one phase shared across all six periods, changes the M2 scale by at most 0.0135. Independent phases in the six periods increase this maximum to 0.1645. The corresponding 24-h gains are 0.0185 and 0.0319 (Methods; Extended Data Fig. 4). These linear sensitivities quantify the benefit of phase-diverse sampling; they do not measure the disturbance amplitude.

Fitting exact 24-h and 12-h sine/cosine pairs shared across the record gives $A_{M2} = 1.061 \pm 0.125$. Allowing those terms to reset in every operating period gives 0.820 ± 0.200. With M2 phase also free, this resetting model no longer excludes zero in the six-period joint test, although the unit, in-phase response remains compatible. This is the relevant boundary of the lunar interpretation: exact daily freedom competes with astronomical components in finite windows, and cannot itself identify an environmental mechanism.

The predesignated B2 round-trip monitor supplies a measured control on the recorded transfer channel. On 1,519 fully matched summer bins, adding B2 changes the whole-template and grouped-lunar scales by 0.0057 ± 0.0072 and 0.0172 ± 0.0094, respectively, using paired delete-day errors (Extended Data Table 5). The full threshold and model comparisons remain in Methods. These small changes constrain the monitored channel on its overlap; clocks and unmonitored laboratory branches are not covered by that control. No same-site null comparison is available.

Physical scope

The observations test the amplitude and phase of the dominant lunar semidiurnal potential on a coastal–inland clock baseline. Ocean loading is substantial in the prediction, but a separate loading scale, 0.937 ± 0.529, is not resolved. Thus the result tests a load-sensitive combined prediction rather than measuring a Love number or independently determining an ocean-loading amplitude. Longer operation with calibrated transfer branches would strengthen that separation.

The associated frequency-ratio preprint by Hou et al. uses overlapping summer observations and reports a whole-template tidal-correlation check.[16] Here the measurement target is the lunar constituent in the longer interrupted record; we do not update the absolute frequency ratio or reuse that preprint's significance. The 2025 commissioning data retain separate within-stage analyses because their conversion and correction conventions differ. None of these short-period fits establishes cross-year level continuity or seasonal potential.

A contemporaneous European fibre-network preprint by Stahl and colleagues compares clocks on approximately 800-km geodetic baselines with a different mixture of diurnal and semidiurnal components, including a transportable clock in one configuration.[21] The observations were acquired independently.

Here the coastal–inland geometry and six operating periods spanning 231 days support a constituent-level test of the lunar response. Both studies include ocean loading; their geographic sensitivities and measurement strategies provide complementary approaches to dynamic chronometric geodesy.

References

1. Ludlow, A. D., Boyd, M. M., Ye, J., Peik, E. & Schmidt, P. O. Optical atomic clocks. *Reviews of Modern Physics* **87**, 637–701 (2015). Source
2. Mehlstäubler, T. E., Grosche, G., Lisdat, C., Schmidt, P. O. & Denker, H. Atomic clocks for geodesy. *Reports on Progress in Physics* **81**, 064401 (2018). Source
3. Chou, C. W., Hume, D. B., Rosenband, T. & Wineland, D. J. Optical Clocks and Relativity. *Science* **329**, 1630–1633 (2010). Source
4. McGrew, W. F. et al. Atomic clock performance enabling geodesy below the centimetre level. *Nature* **564**, 87–90 (2018). Source
5. Bothwell, T. et al. Resolving the gravitational redshift across a millimetre-scale atomic sample. *Nature* **602**, 420–424 (2022). Source
6. Petit, G. & Luzum, B. (eds) IERS Conventions (2010). *IERS Technical Note* **36** (2010). Source
7. LEGOS, NOVELTIS & CLS FES2022 (Finite Element Solution) Ocean Tide. *CNES data set* (2024). Source
8. Farrell, W. E. Deformation of the Earth by surface loads. *Reviews of Geophysics* **10**, 761–797 (1972). Source
9. Han, D. & Wahr, J. The viscoelastic relaxation of a realistically stratified earth, and a further analysis of postglacial rebound. *Geophysical Journal International* **120**, 287–311 (1995). Source
10. Grotti, J. et al. Geodesy and metrology with a transportable optical clock. *Nature Physics* **14**, 437–441 (2018). Source
11. Lisdat, C. et al. A clock network for geodesy and fundamental science. *Nature Communications* **7**, 12443 (2016). Source
12. Riehle, F. Optical clock networks. *Nature Photonics* **11**, 25–31 (2017). Source
13. Voigt, C., Denker, H. & Timmen, L. Time-variable gravity potential components for optical clock comparisons and the definition of international time scales. *Metrologia* **53**, 1365–1383 (2016). Source
14. Denker, H. et al. Geodetic methods to determine the relativistic redshift at the level of 10^{-18} in the context of international timescales: a review and practical results. *Journal of Geodesy* **92**, 487–516 (2018). Source
15. Schröder, S., Stellmer, S. & Kusche, J. Potential and scientific requirements of optical clock networks for validating satellite-derived time-variable gravity data. *Geophysical Journal International* **226**, 764–779 (2021). Source
16. Hou, L. et al. Yb/Sr optical clock comparison at 2.2×10^{-18} uncertainty over a 1350-km fiber link. *arXiv*, 2610.34250v1 (2026). Source
17. Jia, Z.-P. et al. Improved systematic evaluation of a strontium optical clock with uncertainty below 1×10^{-18}. *Metrologia* **63**, 025002 (2026). Source
18. Zhu, Q. et al. Ytterbium lattice clock with systematic uncertainty of 1.3×10^{-18} and instability at the 10^{-19} level. *arXiv*, 2606.10514v2 (2026). Source
19. Chen, F.-X. et al. 10^{-21}-Level optical frequency dissemination over 2067 km of noise-loaded field-deployed fiber network. *Light: Science & Applications* **15**, 276 (2026). Source
20. Schioppo, M. et al. Comparing ultrastable lasers at 7×10^{-17} fractional frequency instability through a 2220 km optical fibre network. *Nature Communications* **13**, 212 (2022). Source
21. Stahl, K. et al. Optical clock network tracks tidal variations of the gravity potential. *arXiv*, 2610.39635v1 (2026). Source

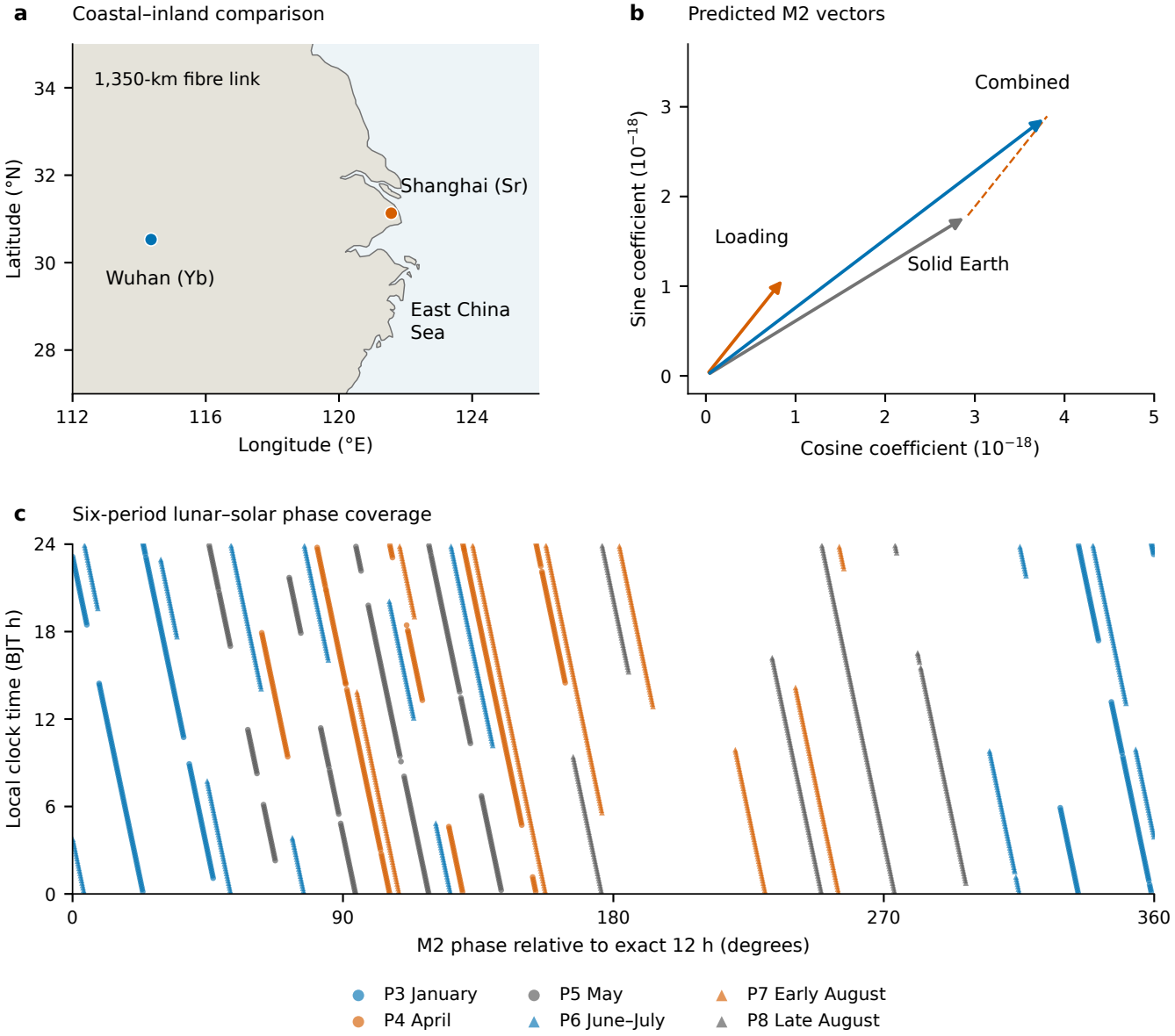


Fig. 1 | Geographic sensitivity and lunar–solar phase coverage. **a,** Nominal Shanghai Sr and Wuhan Yb clock locations relative to the East China Sea; 1,350 km describes the fibre route, without a surveyed route trace. The Natural Earth 1:50M land/coastline map is public domain. **b,** Predicted M2 solid-Earth, loading and combined cosine/sine vectors in one astronomical basis, in 10^{-18} fractional-frequency units before nodal modulation. The dashed segment translates the loading vector. The solid-Earth–loading phase separation is 19.9° and combined amplitude is 4.78×10^{-18}. These are deterministic predictions, without observation-error bars. The full-template loading/total RMS ratio of 34.2% is distinct from the M2 vector-length ratio. **c,** All 2,905 retained 600-s averages, coloured and shaped by six operating periods, shown by local clock time (BJT, UTC+8) and the member-matched astronomical M2 phase relative to the exact 12-h basis, modulo 360°. P3–P8 retain the historical period labels. Each mark represents an average; repeated measurements are not independent replicates. This is a sampling-coverage display, without fitted frequency values, error bars or a hypothesis test. It shows phase diversity, not removal of arbitrary period-specific daily disturbances.

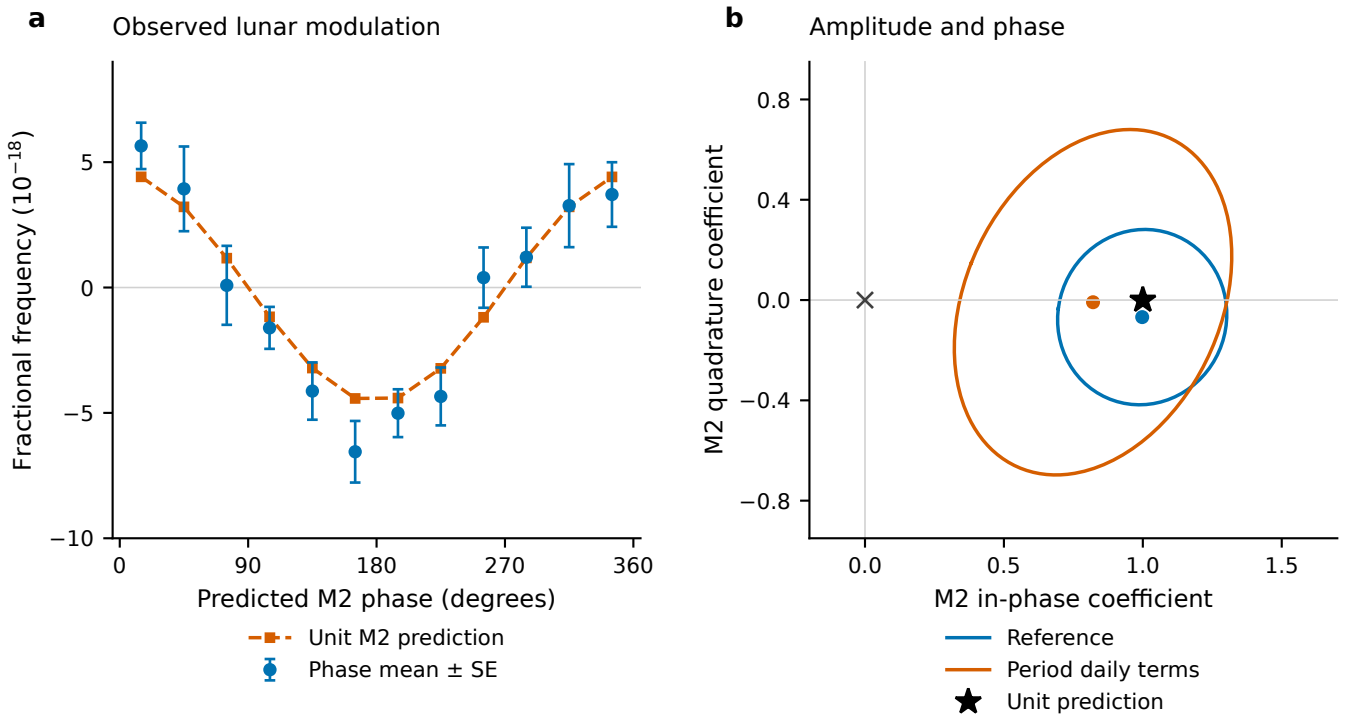


Fig. 2 | Lunar modulation and constituent response. **a,** Means of all 2,905 averages in twelve fixed 30° M2 phase groups after removing only six jointly fitted period constants. Groups contain 222–266 repeated measurements. Bars are one delete-BJT-day jackknife SE with constants refitted in each deletion; orange squares give the unit M2 prediction over the same members. Connecting lines guide the eye. No other periodic term is subtracted, and no segment is separately recentered. The full scatter appears in Extended Data Fig. 2. **b,** Separate-M2 in-phase/quadrature coefficients without daily terms and with period-specific daily terms, with N2 fitted independently. Curves are nominal 95% delete-day F regions with 34 denominator degrees of freedom; the star is the unit prediction and the cross is zero. Both day-based curves exclude zero, whereas the period-level joint tests give P=0.03125 and 0.125, respectively. The two panels reuse the same observations and are not independent tests; phase-group errors are correlated, and day regions do not substitute for period-level inference.

Methods

Physical quantities, nuisance roles and model review

The observable is $y = (R/R_{\mathrm{ref}} - 1)10^{18}$, with $R = f_{\mathrm{Yb,WUH}}/f_{\mathrm{Sr,SHA}}$ and $R_{\mathrm{ref}} = 1.2075070393433377166$. Positive y means that Wuhan Yb is faster relative to Shanghai Sr. For positive Newtonian potential U, the fractional gravitational contribution is $-[\delta U_{\mathrm{WUH}} - \delta U_{\mathrm{SHA}}]/c^2$; templates are converted to the same 10^{-18} working units as y before regression. A template scale tests a specified prediction; it is not identified with an independently measured Love number or with a stochastic trajectory.

Tidal scales and constant level offsets are estimated jointly. Physical scales, calibration nuisance terms and empirical sensitivity terms have separate roles. Offset indicators represent unknown constant levels within recorded periods, groups or segments, conditional on within-group constancy. Daily sinusoids define empirical sensitivity models and are not interpreted as measured environmental states.

The reference design was specified after earlier model comparisons. All twenty declared reference, component and sensitivity designs are retained; no model was selected by proximity to unity or smallest probability. The reference uses the six recorded 2026 operating periods. Group, segment and fixed-calendar partitions test its dependence on level resolution. A statistical nuisance parameter is not thereby an observed physical process.

Frequency conversion and frozen observations

The chain uses $N_{1397} = 858456$, $N_{1550} = 773598$, $N_{1156} = 1295739$ and $N_{1550,\mathrm{WH}} = 966996$. The recorded 2025 relationship gives

$$\alpha = \frac{10.93629485}{25N_{1397}}, \qquad f_{1397} = \frac{f_{\mathrm{Sr}}}{2(1-\alpha)}, \qquad f_{1550,\mathrm{WH}} = \left(N_{1550} + \frac{8}{25}\right)\frac{f_{1397}}{N_{1397}}.$$

The eight reference units include seven on transmission and one at Wuhan. For recorded counter value q in nominal hertz realized by the Wuhan reference,

$$R_{\mathrm{lock}} = \frac{N_{1156}}{N_{1397}} \frac{N_{1550} + 8/25}{N_{1550,\mathrm{WH}} + 26/20 + dq/(20 \times 10^7)} \frac{1}{1-\alpha}.$$

The inherited 2026 configuration uses $d = 1$ and $\alpha = 0$, whereas the separately analysed 2025 records use $d = 2$ and the stated α. For signed Sr and Yb corrections a, b, the reconstruction uses $R = R_{\mathrm{lock}}(1+a)/(1+b)(1+\Delta G)$ with inherited $\Delta G = -3.116 \times 10^{-15}$. Decimal anchors precede conversion to floating point; full ratios are not subtracted in float64.

The corrected experimental intervals and existing theory-independent screen are frozen. Timestamp selection uses the recorded start inclusively and the recorded end plus one second exclusively. Repeated

source identities at shared interval endpoints are retained once; repeated timestamp labels from distinct rows remain. Finite beat offsets within 10^6 Hz of the nominal anchor define each interval's median; rows must lie strictly within 10 Hz of that median. The parent selection retains 1,966,908 rows at this stage.

Screening then forms 300-s blocks from the first timestamp surviving that gross screen in each effective interval. Block means are compared with a 13-block running median, with reflected boundaries and empty blocks treated as missing. The scale is the median absolute deviation of finite block residuals about their median, divided by 0.6744897501960817. Blocks whose absolute residual exceeds six scales and their immediately adjacent blocks are removed. No tidal residual enters this rule. For final averaging, the separate origin is the floored earliest selected timestamp in the interval before quality screening. Half-open, non-overlapping 600-s bins require at least 300 surviving members, including at interval ends; shorter and empty bins are omitted. Observations and interpolated theory are averaged over identical retained source rows.

The 2026 subset contains 1,737,073 members in 2,905 averages. Observations and theory have identical source membership. A separate raw-counter readback previously checked these means, and its identities and SHA256 records accompany this revision. The new fit reads that independently reconstructed response, not any fit-adjusted series. Raw-source readback verifies implementation of the specified conversion and aggregation, conditional on the inherited frequency-chain constants and correction ledger. It covered the selected 2026 members in 34 contributing files, not an independent certification of gate duration, synchronization, selection or correction values. BJT labels are numerical time coordinates; astronomical evaluation applies the documented UTC conversion. Exposure is counter-row count times the nominal one-second gate, including repeated timestamp labels. The 600-s averages contain 300–602 members; member count is neither unique physical seconds nor statistically independent sample size.

The applied 2026 correction values and missing entries are listed in Extended Data Table 1. The ledger uses a signed optical-offset convention: the Sr contribution enters the ratio through $1 + a$, and the Yb contribution through $1 + b$ in the denominator. The table records what was applied computationally; it does not establish whether each shift was already compensated online, its calibrated optical node, or its constancy within an operating period. Unknown clock corrections remain unknown rather than measured zero. Early Sr AC entries and the May Yb correction lack complete calibration. A 37.2-mHz feedback entry at the assumed 698-nm optical node corresponds to 86.6119×10^{-18} for the affected historical ledger entries; their mapping and optical feedback node are not newly certified here. A constant contribution is absorbed by an intercept, but time dependence is not. The historical Yb correction near -1.1×10^{-17} is not added again without resolving overlap with the updated ledger. No correction, mask or observation was changed to improve this revision's agreement. Synthetic period-constant shifts of $(-11, 86.6119, 11, -25, 50, -50) \times 10^{-18}$ change the whole-template and grouped lunar scales by less than 10^{-14} in this design. This checks additive intercept invariance, not the physical accuracy

or inclusion of those corrections. In contrast, adding 0.1 times the target template changes its scale by 0.1: a target-aligned unmonitored variation is inseparable from response in a single channel. These synthetic checks do not bound actual time-dependent or multiplicative errors.

Deterministic tidal synthesis and its limits

The frozen theoretical series combines an IERS-based solid-Earth calculation and FES2022 loading inputs.[6,7,22] Pysolid radial displacements are converted to fractional frequency through

$$s(t) = -\frac{1 + k_2 - h_2}{h_2} \frac{g[u_{\mathrm{WUH}}(t) - u_{\mathrm{SHA}}(t)]}{c^2},$$

with $h_2 = 0.6078$, $k_2 = 0.3010$ and $g = 9.794\ \mathrm{m\,s^{-2}}$. This is a displacement-to-potential approximation, not a complete degree- and frequency-dependent potential calculation. The archived loading construction includes direct load potential, induced potential and displacement through load-Love responses.[8,9] The eighteen-constituent loading input does not supply a complete long-period load model; atmospheric loading has not been independently modelled or bounded. The documented station coordinates are theory-input coordinates rather than newly certified clock-transition locations.

The complete 360,001-point, 90-s theory grid, from 1 September 2025 to 11 September 2026 in the recorded BJT coordinate, is decomposed by equally weighted least squares into quadratures of Q1, O1, P1, K1, N2, M2, S2, K2, Mf, Mm, Msf, Sa and Ssa plus a constant, retaining the remainder. Astronomy uses the recorded FES convention and TT−UTC=69.184 s; Beijing time is converted to UTC once.[23] Decomposition is applied to the deterministic theory, not learned from clock residuals. Linear interpolation on that grid precedes source-member averaging, without extrapolation. The full model scales the total series. Component models jointly fit M2+N2, S2+K2 and K1+O1+P1+Q1, with the remaining prediction subtracted exactly once.

Earlier DE430 geometric-potential and finite-basis frequency-response comparisons supply named sensitivity checks, not certification of all template errors.[24] Their published numerical shifts belong to their original designs and are not copied as uncertainty bounds for the new period reference. The reported template scale remains conditional on the frozen physical construction. The frozen loading phase and sign convention is retained without amplitude or phase adjustment against the observations.

Loading sensitivity and the historical estimator

For the same 2,905 member-matched observations, an exploratory physical decomposition uses $y_i = A_{\mathrm{E}}E_i + A_{\mathrm{O}}O_i + c_{p(i)} + \epsilon_i$, where E is the solid-Earth template and O the loading template. Their sum agrees with the frozen total to 1.3×10^{-32} in fractional frequency. Globally demeaned template correlation is 0.680. After removing period constants, the correlation is 0.686 and 52.9% of the already

period-projected loading-template energy remains after removing the solid-Earth direction. Neither quantity is an observed loading fraction or a measured improvement relative to other baselines.

The estimates are $A_{\mathrm{E}} = 0.903 \pm 0.305$ and $A_{\mathrm{O}} = 0.937 \pm 0.529$, using maximum marginal working errors. The loading-zero conditional test leaves the solid-Earth scale free and gives $P = 0.1875$; the joint unit-point test gives $P = 0.8125$. The latter does not establish separate loading detection. The same six-period assumptions and selection limitations described below apply. An independent SVD implementation checks coefficients, covariances and these tests; this is numerical validation, not independent physical calibration.

The historical summer response used within-segment demeaning, 1,200-s triangular inference windows and precision-weighted combination of segment amplitudes.[16] The archived public-model chain already included a dominant degree-2 solid-Earth tide with real Love numbers and an eight-constituent, displacement-based loading approximation; separate solid-only diagnostics also exist. The present analysis resolves physical frequency groups and tests cross-period amplitude transfer using a different loading construction, while maintaining the provenance of each result. The archived configuration is not assigned wholesale to every earlier developmental calculation.

Resolution of the legacy loading synthesis

The archived loading series is a potential-based, eighteen-constituent calculation, rather than the eight-constituent displacement approximation in the older frequency-ratio work. Its station-potential file uses FES2022b non-extrapolated ocean-height grids and a real load-response kernel proportional to $1+k'_n-h'_n$. The archived prescription uses the Han–Wahr PREM table through degree 696, an extension to degree 40,000, a cosine taper over the last 30% and a Kummer tail. The stated coordinates are Wuhan $(30.529^\circ\mathrm{N}, 114.357^\circ\mathrm{E})$ and Shanghai $(31.130^\circ\mathrm{N}, 121.570^\circ\mathrm{E})$. These are documented model-input coordinates, not newly surveyed clock-transition points.

Let P_j denote a station load-potential phasor formed using the ocean-height lag convention $\exp(-iG)$, and let $D = (P_{\mathrm{SHA}} - P_{\mathrm{WUH}})/c^2$. With the astronomical argument $\theta = V+u$, the consistent time synthesis is $f\,\Re[D\exp(i\theta)]$, equivalently $f\,\Re[D^*\exp(-i\theta)]$. Conjugating the coefficient without reversing the exponential changes the physical phase. The legacy stored series is reproduced over all 360,000 grid points by the latter coefficient paired with the former exponential, $\Re[D^*\exp(i\theta_{\mathrm{old}})]$, omitting f. Here θ_{old} uses the existing Schureman helper at the beginning of each 60,000-point block and advances at the source's fixed constituent speed. The reconstruction RMS is 1.55×10^{-31} in fractional frequency; an independent real-trigonometric implementation gives 1.56×10^{-31}. The original shell command is not needed for this numerical diagnosis and has not been recovered. The archived source's nodal-factor function explicitly returns one.

Projecting both total series on the same thirteen-constituent astronomical basis gives a legacy/current

M2 amplitude ratio 0.81515 and lag difference -27.7367°. The current solid-Earth, loading and total vectors are $(2.9212, 1.7886)$, $(0.8843, 1.1063)$ and $(3.8055, 2.8949)$ in 10^{-18} cosine/sine units. The legacy loading vector is $(0.9227, -1.1437)$. Extended Data Fig. 1c,d shows the current and retired loading vectors side by side. A complete eighteen-loading-plus-five-long-period basis gives a total amplitude ratio 0.81537 and lag difference -27.7557°, showing that omitted minor constituents in the projection do not explain the discrepancy. On a fixed 8,760-hour calendar, the independent restricted PyFES predictor agrees with the current M2 loading argument to 0.0141° and with its waveform to 3.36×10^{-22} maximum fractional frequency; small astronomical-polynomial differences remain. This temporal check shares the station phasors and does not certify their spatial convolution.

The legacy series is therefore excluded because its temporal synthesis is inconsistent with the adopted ocean-lag convention and nodal treatment. Its 0.179 change in fitted M2 scale is not assigned as uncertainty among valid physical templates. The file, its previous results and the source diagnosis remain archived. No clock value, data mask or physical coefficient was changed to obtain the exclusion, and no optimized phase or time shift was used in the reconstruction.

For completeness, a separate data diagnostic holds N2, near-12-h and diurnal templates, the remainder and all nuisance levels fixed while comparing the two M2 predictions in the current basis. The legacy unit point is $(0.7215, -0.3794)$; the measured pair is $(0.9977, -0.0680)$. Its six-period conditional test gives $P = 0.0625$, versus 0.96875 at the current unit point. If amplitude along each direction is free, the legacy-phase test gives $P = 0.09375$. The approximate delete-day ellipse excludes the legacy unit point, whereas these period tests do not at 5%. Accordingly, exclusion of the legacy synthesis rests on the source and numerical construction, not a claim that the data alone reject its phase. Earlier phase pairs obtained after replacing all other template components used a different nuisance model and are not directly comparable.

Joint estimation and uncertainty

All fits use equal weights for the retained 600-s averages, consistent with this reduction route. Columns are normalized for rank checks and QR solution; reported coefficients use their original units. SVD supplies an independently implemented numerical check. Rank loss or numerical near-degeneracy is reported rather than resolved with an arbitrary physical amplitude: the smallest singular value of the column-normalized design must exceed 10^{-11} times the largest. Zero nuisance columns after deletion are removed; loss of a target direction is an error.

For $y = L\theta + Z\gamma + \epsilon$, partial regression is implemented with both $M_Z y$ and $M_Z L$, where $M_Z = I - ZZ^+$. This is algebraically equivalent to the full joint fit.[25] Its noise covariance is $M_Z \Sigma M_Z^\mathsf{T}$; a residual is not a new independent observation. Least-squares covariance uses the full-model residual degrees of freedom. BJT-day and operating-period jackknives refit all retained parameters. For G deletion groups,

$$C_{\mathrm{JK}} = \frac{G-1}{G}\sum_{g}(\hat{\theta}_{-g}-\bar{\theta})(\hat{\theta}_{-g}-\bar{\theta})^{\mathsf{T}}.$$

We retain LS, delete-day and delete-period covariances separately.[26] The largest marginal standard error is a working summary, not a joint covariance or a calibrated confidence interval. The phase contours use delete-day covariance and $2F_{2,34}(0.95)$; their coverage is conditional and is not certified by the covariance ellipse alone.

Restricted wild diagnostics refit each candidate null, enumerate all $2^6 = 64$ operating-period sign patterns and recompute the delete-period studentizer for each realization.[27] All patterns are retained, including globally positive and negative ties. The two-sided minimum is $2/64 = 0.03125$. The procedure assumes approximately independent, sign-symmetric period errors; nuisance estimation prevents interpreting it as an exact experimental randomization test. It does not adjust for exploratory model selection. Scalar and joint-phase probabilities are distinct. No new confidence set is inferred from tests at zero and unity alone.

Independent scaled-SVD calculations check the twenty declared designs, including coefficients, covariances and both null tests. Joint and projected solutions agree numerically; eight fixed-calendar designs also reproduce the archived calculations. These are implementation checks, not physical calibration or arbitrary-noise coverage guarantees. The complete test ledger is retained with the code.

Validation and subtraction of a dominant component

A physical analysis may first examine the dominant constituent and then ask whether weaker predictions are supported. Agreement of a strong component with its expected amplitude and phase supports that component's model; it neither certifies all frequencies nor supplies an independent, error-free calibration. A fit to the whole template already contains the weak constituents and is not an independent strong-only subtraction. Constituent membership must be explicit, with each fixed remainder included once.

For a same-data decomposition $y = Sa + Wb + Zc + \epsilon$, let $s = M_Z S$, $w = M_Z W$ and $v = M_Z y$. The strong-first calculation uses

$$w_{\perp} = w - s(s^{\mathsf{T}}s)^{-1}s^{\mathsf{T}}w, \qquad v_{\perp} = v - s(s^{\mathsf{T}}s)^{-1}s^{\mathsf{T}}v,$$

$$\hat{b} = (w_{\perp}^{\mathsf{T}}w_{\perp})^{-1}w_{\perp}^{\mathsf{T}}v_{\perp}.$$

The strong and calibration coefficients are then recovered by back-substitution. This is block elimination

of the joint model, not freezing the strong-only fit after a significance screen. The complete linear map from the original observations propagates covariance; every day/period deletion repeats all stages. If the strong component instead comes from independent calibration, its calibration uncertainty must enter the residual covariance; such an independent amplitude/phase calibration is not assumed here.[25]

We checked this construction by separating the member-matched M2 and N2 columns, keeping the near-12-h and diurnal scales, fixed remainder and calibration design unchanged. Their demeaned template RMS values are 3.254 and 0.686×10^{-18}. In the period-offset design, their correlation after removing the common nuisance space is 0.06880; projecting M2 then retains 99.527% of the already nuisance-projected N2 energy. This fraction is not measured relative to the original, unprojected N2 template. Subtracting only the strong-only fit from the response and regressing on the unprojected weak column would attenuate the weak coefficient by $1 - \rho^2$ in this two-column setting.

Both elimination orders and joint QR agree with independently checked SVD operators; full covariance differences are below 2.3×10^{-12} in the calculation units. The M2/N2 emphasis was chosen retrospectively after the grouped-lunar analysis; there is no multiplicity or model-selection adjustment. Both estimators remain reported.

The separate M2 phase targets are its in-phase and quadrature coefficients; N2 retains an independent fixed-phase scale. Near-12-h and diurnal scales remain free. If the inherited M2 template is $s = a\cos\theta + b\sin\theta$, its quadrature is $q = a\sin\theta - b\cos\theta$, with the same nodal basis and source-member averaging. This represents a 90-degree phase rotation, not orthogonality under the observed sampling. The corresponding identities are checked directly against the frozen columns.

Without daily terms, the two M2 coefficients are $(0.998, -0.068)$ with marginal working errors $(0.123, 0.177)$; the joint zero and unit-point probabilities are 0.03125 and 0.96875. With period-specific daily terms, they are $(0.821, -0.009)$ with errors $(0.198, 0.285)$ and probabilities 0.125 and 0.6875. These component tests use the same observations; their emphasis in the present narrative does not retrospectively preregister them or adjust for model selection. Their SVD check independently verifies coefficients and LS covariance; the deletion and sign tests use the validated shared library. High unit-point acceptance is not a probability that the physical prediction is true or a test of equivalence.

M2 phase folding and summer support

The M2 display uses $\phi_i = \mathrm{atan2}(Q_{M2,i}, T_{M2,i})$ modulo 2π, evaluated from the member-matched quadratures. Its origin is the predicted M2 maximum, not an estimated phase shift. Twelve equal phase intervals were fixed before rendering this check. Displayed observations are $y_i - \hat{c}_{p(i)}$, where the constants come from the simultaneous separate-M2/N2 fit; the fixed remainder and other periodic terms are not subtracted from this display. For each deleted BJT day, all joint-fit coefficients are re-estimated and

the retained observations are regrouped using the unchanged phase intervals. The jackknife of these means gives the plotted working errors. Individual observations remain equally weighted, as in the fitted estimator. The same data determine the constants and the phase means; their dependence is retained in this deletion calculation. The plot is not a new measurement of M2 amplitude or a separate significance test.

The summer restriction is fixed to P6–P8. Its separate M2 scale uses exactly the same design roles and maximum-error rule as the full six-period fit, with only the observations and corresponding period indicators restricted. It does not inherit the full-record period-test calibration, and no three-cluster tail probability is quoted. Independent SVD and partial-regression checks reproduce the full and summer coefficients and deletion covariances.

Solar-day tests, display and transfer

Daily sensitivity columns contain exact 24-h and 12-h sine/cosine pairs, globally or separately by operating period or occupied calendar block. They make no independent claim about the disturbance's mechanism or measured amplitude. S2 and a free 12-h sinusoid share a direction; binding S2 to K2 may leave a nonzero group direction without separately identifying S2. Near-degeneracy is evaluated through projected target information as well as the whole-design condition number.

For the analytic solar-disturbance calculation, let h_L be the row of the fixed reference fit operator that maps observations in units of 10^{-18} to A_L, and let Q_τ contain the source-member-averaged cosine and sine at period τ. A unit-amplitude sinusoid has a phase vector of Euclidean norm one. Its largest absolute effect is

$$G_{\text{shared}}(\tau) = \|h_L Q_\tau\|_2, \qquad G_{\text{separate}}(\tau) = \sum_p \|h_L I_p Q_\tau\|_2,$$

where I_p selects operating period p. The second bound is attained by choosing each phase to align its projected contribution; it assumes the same prescribed amplitude in every period. For amplitude $a \times 10^{-18}$ the bounds scale by $|a|$. An independent scaled-QR solve with the extremizing perturbations reproduced the bounds to 2×10^{-15} in A_L. This is a deterministic geometry check, not an injection-based noise calibration. Both the nuisance coefficients and other fitted tidal bands respond in the joint fit. These gains describe the reference estimator without free daily pairs; they must not be substituted for the estimates obtained after adding those pairs.

For a separate retrospective transfer check, the grouping was declared before this calculation: January/April/May (1,263 bins) versus June–July/early August/late August (1,642 bins). Only the target amplitude is transferred. Test-set nuisance coefficients are re-estimated under both zero and learned amplitude, including other tidal groups in lunar fits. The score is $100(RSS_0 - RSS_A)/RSS_0$, on iden-

tical test observations and nuisance spaces. All eight comparisons are reported (Extended Data Table 3). They are conditional predictive scores, not P values or independent replications, and do not connect absolute period levels. Different designs have different null RSS; improvements are not used to select a model. Independent scaled-SVD calculations reproduce all eight amplitudes and scores. Period norm shares normalize full-record nuisance-projected template energy; test-set target norms instead use a local projection. Neither is a noise-calibrated measure of independent information.

Figure 3a chooses the 48-h window with the largest retained source-member count, irrespective of fit residuals. The amplitude is learned from other periods; one nuisance level is fitted using the held-out period. That level is subtracted for display only, with no separate segment recentering. Eighteen-bin display means never cross effective-interval boundaries or gaps exceeding 900 s. No plotted residual, phase contour or goodness-of-fit change is counted as an additional independent detection.

Held-out scores compare the same observations and locally fitted nuisance design with zero amplitude or an amplitude learned from other periods. The period-offset design gives a 3.525% pooled RSS reduction; period-daily freedom gives 0.539%, improving four of six periods. The shared training data make these conditional transfer checks, rather than six independent replications.

Conditional checks of the six-period test

The reported marginal error and the studentizer in the period-level test are different constructions. For the period-daily lunar scalar model, the working error is 0.20548, whereas its delete-period studentizing error is 0.11135. At unit response, the observed squared statistic is 4.61815; the largest of the 62 non-uniform sign-pattern statistics is 3.59197. Only the two globally uniform patterns reach the observed value, producing $P = 2/64$. This explains the enumeration without interpreting the working error as an inverted test interval. The restricted nuisance fit differs between zero and unit nulls.

We examined the estimator on its frozen sampling under three explicitly synthetic noise scenarios: independent Gaussian noise with standard deviation 20×10^{-18}; independent white noise with standard deviation 10×10^{-18} plus an Ornstein–Uhlenbeck process with standard deviation $\sqrt{300} \times 10^{-18}$ and 12-h correlation time; and variance-normalized Student-t_7 innovations with MA(1) coefficient -0.15 and marginal standard deviation 20×10^{-18}. MA memory restarts at effective-interval changes or gaps of at least 900 s. These parameters define test scenarios and are not claims about measured noise.

Each scenario uses 512 realizations, seed 2026110501, with true in-phase response one, quadrature zero and other nuisance coefficients zero. The last choice follows linear equivariance conditional on fixed designs and identifiable full and deletion fits, not an assertion that experimental offsets vanish. The fixed theoretical remainder has already been removed in this simulation coordinate, as in the estimator. The same simulated errors are used across the three declared designs. Every realization retains all 64 restricted sign patterns and recomputes the period studentizer. Vectorized and directly enumerated P

values agreed for the first realization of each design/scenario. The OU process evolves across actual elapsed time, including inter-period gaps, rather than being reset to enforce exact period independence.

True-value acceptance at the 5% threshold ranges from 95.12% to 97.46% (Extended Data Table 6). Monte Carlo standard errors are approximately 0.70–0.95 percentage points. These finite scenarios do not indicate pronounced undercoverage, but they do not certify non-stationary experimental noise, unmodelled corrections, template uncertainty or selection-adjusted inference. Marginal max-error intervals and joint-phase acceptance are distinct quantities. No observation, reference model or nuisance parameterization was changed following these tests. Additional simulations assess the whole-template scalar and two M2 phase designs separately, using 512 realizations per scenario and seed 2026110503. True-unit acceptance is 94.92%–96.88%, with Monte Carlo standard errors of 0.77–0.97 percentage points. These checks neither replace nor pool the grouped-lunar tests (Extended Data Table 6). Errors are shared within each three-design batch, so its nine table rows do not represent nine independent simulated experiments.

The earlier grouped-lunar delete-day F contours were also tested, separately from the wild procedure. With seed 2026110504, 512 realizations per declared noise scenario and the unchanged threshold $2F_{2,34}(0.95)$, joint true-point coverage ranges from 94.53% to 99.41% (Extended Data Table 6). These finite scenarios test that grouped-lunar construction; their numerical coverage is not assigned to the separate-M2 contours in the present Fig. 2b. Experimental tail probabilities, unmodelled systematics and selection-adjusted coverage are not calibrated by these checks. The maximum marginal SE, a coordinate interval, a day-based joint contour and a period-studentized test remain distinct summaries.

Experimental controls and longer time scales

The B2 diagnostic reuses archived unique member matching without interpolation and requires complete support for every retained bin. The 10, 100 and 1,000 Hz gross-excursion thresholds were introduced after monitor inspection in the earlier analysis; they are not preregistered experimental criteria. All are retained here. The 100 Hz configuration contains 1,519 bins, 910,041 nominal members and 20 BJT days, only from P6–P8. Frozen indices, times, member counts and period identities match the current independently reconstructed observations. The matching algorithm and hardware synchronization were not independently rerun or certified in this check.

Both the baseline and B2-augmented models are fitted on that same support. With 100 Hz screening and no daily terms, the whole-template scale changes from 0.66817 to 0.67390; the lunar scale changes from 0.82353 to 0.84074. These subset estimates are not the six-period reference. Paired deletion errors use the difference of the two fit operators, retaining their dependence. Day and three-period deletion errors are separately reported (Extended Data Table 5), without assigning calibrated confidence to three clusters. Adding daily terms changes the two scales by 0.0402 and 0.0393. The intermediate

threshold was inherited, not chosen to minimize these changes.

B2 enters as a freely scaled empirical regressor. Its fitted coefficient is an association parameter, not a calibrated round-trip-to-clock transfer function. Its separate template projection uses optical normalization 193399200000000 Hz without a one-way factor of two. An arbitrary nonzero rescaling of a freely fitted B2 column leaves its regression space unchanged; a small change in the clock scale therefore does not validate the normalization or exclude tidal-frequency content in the monitor. Increasing the threshold from 100 to 1,000 Hz adds only one bin yet substantially changes the monitor projection. All thresholds are reported, without assigning a hardware cause to that bin or treating a smaller response shift as stronger exclusion. The monitor constrains only its measured channel and overlap, not clocks, unmonitored branches or earlier periods.

No long-term stability estimate obtained after fit-based offset adjustment is included. Short-period offset models cannot infer seasonal levels they have removed. Previously explored seasonal bounds use a separate level-preserving model and incomplete long-period loading; they are not new detections or limits from the present reference fit.

Independent processing comparison

We compared the present analysis with a separately developed period-offset reduction of the same experiment, frozen at its R3 source release (Extended Data Table 3). All 42 raw counter files used in that reduction matched its recorded SHA256 identities. A separate parser reconstructed its 3,299 retained bins, 1,965,327 bin members, frequency-ratio coordinate and eight period-noise weights. Scaled-SVD calculations independently reproduced its nine reported fits, deletion errors and correlations to 5×10^{-14} in their reported numerical coordinates. A separate reconstruction also reproduced the 20 present model designs, deletion covariances and all 64-pattern test probabilities. These checks verify computations conditional on the recorded inputs and conventions; they are not independent clock experiments.

The two reductions differ in support and averaging. The R3 route floors time labels to seconds, uses fixed 600-s grid centres with at least 180 retained members, evaluates the template at the centre and assigns bin standard deviations $\sigma_i = \alpha_{p(i)}/\sqrt{n_i}$. The fit uses inverse-variance weights $w_i = 1/\sigma_i^2$. The present route uses the retained member times and the same averaging operator for observations and predictions, with equal bin weights. Among 2026 members, 1,736,236 source-file/line identities are shared; 837 occur only in the present reduction and 10,844 only in R3. The resulting 2,905 and 2,933 averages are not interchangeable observations. On the common theoretical time grid, the largest difference between the two current combined predictions is 9.37×10^{-22} in fractional frequency. These numerical reconstructions share physical inputs and the astronomical library; they verify the consumed-input calculations, rather than independently certifying the loading convolution, clock coordinates or

deformation-to-potential approximation. The legacy stored-series temporal construction is diagnosed above and excluded from the adopted prediction.

The R3 eight-period lunar estimate is 0.9866 ± 0.1231, where its quoted error is max(scaled LS, delete-UTC-day). Restricting that route to its six 2026 periods gives 0.9899 ± 0.1255; changing only its bin weighting to equal weights gives 0.9538 ± 0.1275. This weighting contrast isolates one change within R3. It does not attribute the cross-pipeline difference, which also involves membership, bin timing, response reconstruction and template averaging. Agreement checks implementation and processing sensitivity on overlapping observations; it does not independently calibrate the physical inputs or the estimated noise weights. The estimates are not pooled or treated as independent replications. The 2025 relative-only reduction is not evidence for calibrated cross-year levels. Neither its errors nor the present conditional test probabilities or simulation coverage are transferred between estimators. The present six-period, member-matched reference remains unchanged.

Additional model and transfer diagnostics

The whole-template six-period scale is 0.913 ± 0.105; its conditional zero and unit tests give $P = 0.03125$ and 0.5625. Group and segment constants give 0.885 ± 0.109 and 0.832 ± 0.120. The grouped lunar scale is 0.973 ± 0.121, with near-12-h and diurnal scales 1.005 ± 0.271 and 0.462 ± 0.415. Group and segment constants give lunar scales 0.945 ± 0.123 and 0.952 ± 0.137. The phase pair for grouped lunar response is $(0.972, -0.070)$, with joint zero/unit probabilities 0.03125/0.90625. All twenty earlier designs are retained in Extended Data Table 2.

Shared and period-specific daily terms give grouped-lunar scales 1.023 ± 0.123 and 0.761 ± 0.205. The period-daily design retains 38.6% of the period-reference nuisance-projected lunar-template squared norm. Ten-day and five-day daily resets give 0.891 ± 0.282 and 0.893 ± 0.430, with scalar period probabilities 0.0625 and 0.09375 at zero. The scalar period-daily probability is 0.03125 at both zero and unity; its joint-phase unit test instead gives 0.125. A maximum marginal SE is not the inversion of these conditional tests.

For separate M2, period-daily freedom gives a fixed-phase scalar zero-test probability 0.03125, but the two-parameter amplitude–phase zero test gives 0.125. The abstract's statement that zero is not excluded refers specifically to the latter. The joint unit-point probability is 0.6875. Both coordinates in Fig. 2b use the same data; the nominal 95% contours use delete-day covariance and $2F_{2,34}(0.95)$, not inversion of the six-period test.

For the M2 estimator, prescribed unit-amplitude daily sinusoids give maximum shared/reset-period gains of 0.013527/0.164485 at 12 h and 0.018464/0.031907 at 24 h, per 10^{-18} fractional frequency. The corresponding grouped-lunar gains are 0.010430/0.158707 and 0.016347/0.031691, respectively (Extended Data Fig. 4). No environmental amplitude is inferred by this operator calculation.

Leave-one-period-out whole-template training scales span 0.864–1.002. Test RSS improves in five of six periods; the sixth changes by less than 0.02%. Pooled RSS decreases by 3.53%. These are conditional prediction checks with locally fitted test-period levels and overlapping training sets, not independent detections (Extended Data Fig. 3). For the retrospective early/summer split, separate M2 estimates are 1.063 ± 0.487 and 0.882 ± 0.188, while whole-template point estimates are 1.087 and 0.704. The differences motivate a constituent-specific narrative but are not a test of equal responses. The component emphasis remains retrospective.

Deterministic tides are predictions; constant scales describe response to those predictions; offsets represent constant calibration nuisance; exact daily sinusoids define empirical sensitivity classes; residual covariance describes unexplained variation; quadratures test constant phase. None of these fitted objects alone is an observed environmental state or an independently measured calibration.

Reproducibility and availability

The reproducibility package contains frozen observations, source-member identities, all declared model results, verification records and figure-generation code. Numerical reproduction starts from the derived table, with the preceding independent raw-counter readback documented separately. The analyses use the same experimental observations; computationally independent implementations are not independent experiments.

Data availability

The author-review package includes the 2,905 derived frequency averages, member-matched theory columns, source-data tables and verification records needed to assess the principal results. The underlying counter-source manifest and independent readback are documented separately. The package is prepared for provision to editors and referees with submission; a public repository record and persistent identifier have not yet been assigned.

Code availability

A versioned review archive contains executable code for the principal frozen-bin estimates, error construction, phase folding and source-data checks, with environment and input hashes. Its portable reproduction entry has been executed. The original processing and physical-template source paths are recorded separately. A public code archive, license and persistent identifier must be finalized before publication; no public repository or DOI is claimed in this draft.

Assistance disclosure

OpenAI Codex assisted numerical implementation, code review, figure generation and drafting. Authors remain responsible for scientific verification, authorship, disclosures and access statements. Automated

reviews are not author approval.

Methods references

22. Lyard, F. H., Allain, D. J., Cancet, M., Carrère, L. & Picot, N. FES2014 global ocean tide atlas: design and performance. *Ocean Science* **17**, 615–649 (2021). Source
23. Sutterley, T. C. et al. pyTMD: Python-based tidal prediction software, version 3.0.9 (version used in the primary analysis). Source
24. Folkner, W. M., Williams, J. G., Boggs, D. H., Park, R. S. & Kuchynka, P. The planetary and lunar ephemerides DE430 and DE431. *Interplanetary Network Progress Report* **42-196**, 1–81 (2014). Source
25. Ding, P. The Frisch–Waugh–Lovell theorem for standard errors. *arXiv*, 2009.06621 (2020). Source
26. Efron, B. & Stein, C. The Jackknife Estimate of Variance. *The Annals of Statistics* **9**, 586–596 (1981). Source
27. Cameron, A. C., Gelbach, J. B. & Miller, D. L. Bootstrap-Based Improvements for Inference with Clustered Errors. *Review of Economics and Statistics* **90**, 414–427 (2008). Source

Acknowledgements and funding

Author-confirmed acknowledgements, funding agencies and award identifiers are required before submission.

Author contributions

Author-specific contributions are to be supplied and approved by all authors.

Competing interests

Author declarations are required before submission.

Additional information

Correspondence and requests for materials should be addressed to the corresponding authors; their names and contact details will be added with the author list. This author-review PDF includes the Article, Methods and Extended Data; no separate Supplementary Information is required for the present version.

Extended Data

Extended Data Table 1 | Retained support by operating period. Members are retained nominal one-second counter rows, including repeated time labels. Hours are member count divided by 3,600, not verified continuous gate uptime or statistical effective sample size. BJT-day counts use bin-mean days. P3–P8 preserve the historical eight-period identifiers; the two 2025 commissioning periods are excluded from this analysis.

Period	Calendar stage	Bins	Members	Nominal hours	Intervals	BJT days
P3	January	403	241,474	67.08	9	4
P4	April	440	261,782	72.72	11	5
P5	May	420	250,768	69.66	13	4
P6	June–July	599	358,311	99.53	8	9
P7	August a	592	355,036	98.62	5	8
P8	August b	451	269,702	74.92	3	5

Applied correction coordinate | Corrections used in the 2026 response coordinate. Values of a and b are in 10^{-18} and enter the ratio as $(1+a)/(1+b)$, after conversion to fractional units. A range gives applied group-specific values, not uncertainty. The Sr entries combine the corrected experimental table's density, Zeeman and available AC terms with the inherited -0.67×10^{-18} term. The common inherited $\Delta G = -3.116 \times 10^{-15}$ is applied separately. Missing terms are omitted from this numerical coordinate rather than asserted to vanish physically.

Period	Applied a	Applied b	Incomplete entries
P3	-93.4981	-9.9 to -9.6	Sr AC
P4	-181.1900	1.8 to 1.9	Sr AC
P5	-181.1000	0.0	Sr AC; Yb (0 is a placeholder)
P6	-172.0400	1.7 to 1.9	No missing Sr table entry; see qualification below
P7	7.9300	1.9 to 2.0	No missing Sr table entry; see qualification below
P8	8.0700	1.9 to 2.1	No missing Sr table entry; see qualification below

For P3, a includes the ledger's positive 37.2-mHz feedback entry converted using an assumed 698-nm optical node; no such entry is applied to P4–P8. The same entry applies to the separately analysed September and October 2025 records. P3–P4 Yb values follow the corrected table; P6–P8 values follow the archived clock-parameter record. June is assigned to the June–July correction period. Having no missing table entry does not certify a complete clock calibration. Independent confirmation of online/offline compensation, actual frequency reference and within-period time dependence remains unavailable. These limitations are distinct from the checked arithmetic and source membership.

Extended Data Table 2 | All declared reference, component and sensitivity estimates. Marginal errors are working standard errors; P values are conditional two-sided six-period diagnostics. Phase rows report the in-phase marginal coefficient but use a joint two-parameter null test. They must not be read as scalar tests. All models use the same 2,905 observations.

Model	Estimate	LS SE	Day SE	Period SE	P(0)	P(unit)
P-W	0.913	0.087	0.101	0.105	0.03125	0.56250
P-L	0.973	0.118	0.121	0.095	0.03125	0.81250
P-Q	0.972	0.118	0.121	0.099	0.03125	0.90625
G-W	0.885	0.090	0.108	0.109	0.03125	0.40625
G-L	0.945	0.119	0.123	0.095	0.03125	0.53125
S-W	0.832	0.097	0.120	0.102	0.03125	0.12500
S-L	0.952	0.127	0.137	0.098	0.03125	0.59375
P-W+DG	1.062	0.111	0.128	0.158	0.03125	0.65625
P-L+DG	1.023	0.119	0.123	0.111	0.03125	0.81250
P-W+DP	0.706	0.174	0.215	0.106	0.03125	0.06250
P-L+DP	0.761	0.190	0.205	0.111	0.03125	0.03125
P-Q+DP	0.754	0.191	0.201	0.100	0.03125	0.12500
10-W	0.884	0.088	0.100	0.097	0.03125	0.31250
10-W+DB	0.790	0.209	0.277	0.129	0.03125	0.28125
10-L+DB	0.891	0.234	0.282	0.188	0.06250	0.56250
10-Q+DB	0.866	0.243	0.277	0.181	0.25000	0.71875
5-W	0.886	0.088	0.101	0.097	0.03125	0.31250
5-W+DB	0.924	0.291	0.383	0.232	0.06250	0.78125
5-L+DB	0.893	0.308	0.430	0.307	0.09375	0.71875
5-Q+DB	0.892	0.308	0.437	0.305	0.31250	0.90625

P, operating-period constants; G, source-group constants; S, valid-segment constants; 10 and 5, calendar-block constants (days). W, whole template; L, grouped M2+N2 scale; Q, grouped-lunar in-phase and quadrature pair. DG, one shared set of 24-h/12-h terms; DP, one set per period; DB, one set per calendar block. Q rows report the in-phase coefficient, but their P values test the joint pair. All numerical entries are unchanged.

Extended Data Table 3 | Bidirectional transfer of an amplitude between three-period groups. Early denotes P3–P5 (1,263 bins) and late P6–P8 (1,642 bins). Only the target amplitude is learned from the training group. Test-set nuisance coefficients are estimated on identical test observations under zero and learned amplitude. Daily denotes period-specific 24-h and 12-h terms. RSS values use the squared 10^{-18} working units; reductions use unrounded values, $100(RSS_0 - RSS_A)/RSS_0$. These retrospective scores are not significance tests.

Model	Train → test	Train A	RSS_0	RSS_A	Reduction (%)
Whole	Early → late	1.087	805576.4	796730.2	1.098
Whole	Late → early	0.704	475418.0	443953.3	6.618
Whole + daily	Early → late	0.539	794135.5	788831.8	0.668
Whole + daily	Late → early	0.782	430625.8	429618.6	0.234
Lunar	Early → late	1.051	803272.2	791400.7	1.478
Lunar	Late → early	0.872	451140.6	439756.4	2.523
Lunar + daily	Early → late	0.542	793651.3	788832.9	0.607
Lunar + daily	Late → early	0.844	430191.2	429590.7	0.140

The early group contains 754,024 retained nominal one-second members and the late group 983,049; these counts do not certify continuous gate uptime. Positive gains do not establish equal amplitudes between groups. Operating-period labels and retained support are listed in Extended Data Table 1.

Independent processing comparison | Processing comparison on the same clock experiment. All rows use jointly fitted period levels and three tidal bands; none shifts individual segment centres to their fitted tides. R3 denotes the separately developed, frozen period-offset reduction. Its last two rows are retrospective calculations for this comparison. Errors are working summaries conditional on fixed inputs; their differing construction prevents direct confidence-level comparison. R3 samples predictions at bin centres, whereas the present reduction averages the prediction over actual retained members. The eight-period result does not establish cross-year level continuity.

Reduction	Bins	Weighting	A_L	LS SE	Day SE	Period SE	Working SE
Present (6)	2,905	Equal	0.9726	0.1183	0.1205	0.0947	0.1205
R3 (8)	3,299	WLS	0.9866	0.1231	0.1116	0.1314	0.1231
R3 (6)	2,933	WLS	0.9899	0.1255	0.1150	0.1251	0.1255
R3 (6)	2,933	Equal	0.9538	0.1275	0.1192	0.1111	0.1275

Parentheses give the number of operating periods; six-period rows use only 2026 observations. For the present reduction, the quoted SE is the maximum of all three columns, using BJT days. For R3 it is the maximum of LS and delete-UTC-day, with delete-period retained as a diagnostic. WLS uses $\sigma_i = \alpha_{p(i)}/\sqrt{n_i}$ and inverse-variance weight $n_i/\alpha_{p(i)}^2$. R3 noise coefficients are estimated after subtracting the unit template within effective intervals and are held fixed during fitting and deletions; their uncertainty is not propagated. No long-term stability is inferred after fit-based level adjustment.

Extended Data Table 4 | Strong/weak constituent cross-check on unchanged observations. M2 and N2 have separate constant scales; near-12-h and diurnal scales are estimated jointly, with the same fixed remainder as the grouped lunar model. Uncertainties are max(LS, delete-day, delete-period) working errors. P values are conditional, two-sided six-period sign diagnostics with minimum 0.03125, without multiplicity or model-selection adjustment. No strong coefficient is fixed after a P-value screen. N2 is not individually resolved in these designs.

Design	M2	N2	M2 P(0)	M2 P(1)	N2 P(0)	N2 P(1)
Period	0.997 ± 0.123	0.541 ± 0.648	0.03125	0.93750	0.34375	0.34375
Segment	0.942 ± 0.135	1.137 ± 0.830	0.03125	0.46875	0.15625	0.87500
Period+daily	0.820 ± 0.200	-0.043 ± 0.852	0.03125	0.28125	0.96875	0.12500

Summer and archived-template checks | Same-estimator summer and historical-template checks. All rows use period constants, equal bin weights and max(LS, delete-BJT-day, delete-period) working SE. Values are not calibrated confidence intervals. The historical row is a retained audit of the now-excluded legacy synthesis; it is not a valid-model uncertainty alternative. The summer subset overlaps the full record; template alternatives also share all observations.

Sample and template	Bins	Whole scale	M2+N2 scale	Separate M2 scale
Six periods, current	2,905	0.913 ± 0.105	0.973 ± 0.121	0.997 ± 0.123
Summer P6–P8, current	1,642	0.704 ± 0.174	0.872 ± 0.180	0.882 ± 0.188
Six periods, excluded legacy	2,905	1.045 ± 0.154	1.138 ± 0.213	1.176 ± 0.200

The corresponding summer M2 phase fit gives $(0.882, 0.018)$ with marginal errors $(0.188, 0.194)$. The full-record historical phase fit gives $(1.166, 0.609)$ with errors $(0.176, 0.201)$. Each phase pair refers to its own template. Phase-fold source data retain all 2,905 individual observations and all twelve group summaries.

The current-basis legacy M2 prediction has amplitude ratio 0.81515 and lag difference −27.7367°. Under identical current nuisance columns its unit-point and free-amplitude phase tests give P=0.0625 and 0.09375. The legacy series is excluded for its diagnosed conjugation/nodal synthesis, not because these tests reject at 5%. The early P3–P5 separate-M2 scale is 1.063 ± 0.487 (1,263 bins).

Extended Data Table 5 | B2 association checks on identical matched observations. W denotes whole template, L grouped lunar, and +d period-specific daily terms. B is the baseline scale. The response change is augmented minus baseline, with paired delete-day SE in parentheses; the next column gives its paired delete-period SE. The final column is the monitor's template projection under optical normalization with max(LS, delete-day, delete-period) SE. These are working errors, not calibrated intervals or physical transfer bounds. Support is P6–P8 only; all three archived gross-excursion screens are shown.

Hz	Model	Bins	B	Change (day SE)	Period SE	B2 projection (max SE)
10	W	1476	0.697	0.0074 (0.0103)	0.0114	0.074 (0.087)
10	W+d	1476	0.736	0.0175 (0.0182)	0.0166	0.173 (0.144)
10	L	1476	0.835	0.0083 (0.0061)	0.0044	0.082 (0.108)
10	L+d	1476	0.823	0.0214 (0.0196)	0.0119	0.210 (0.165)
100	W	1519	0.668	0.0057 (0.0072)	0.0090	0.075 (0.152)
100	W+d	1519	0.726	0.0402 (0.0326)	0.0313	0.516 (0.433)
100	L	1519	0.824	0.0172 (0.0094)	0.0096	0.223 (0.192)
100	L+d	1519	0.820	0.0393 (0.0239)	0.0095	0.502 (0.354)
1000	W	1520	0.667	0.0004 (0.0049)	0.0073	-1.169 (2.592)
1000	W+d	1520	0.724	0.0006 (0.0368)	0.0399	-1.227 (4.390)
1000	L	1520	0.821	0.0019 (0.0141)	0.0067	-3.501 (5.912)
1000	L+d	1520	0.817	0.0019 (0.0340)	0.0231	-3.728 (5.734)

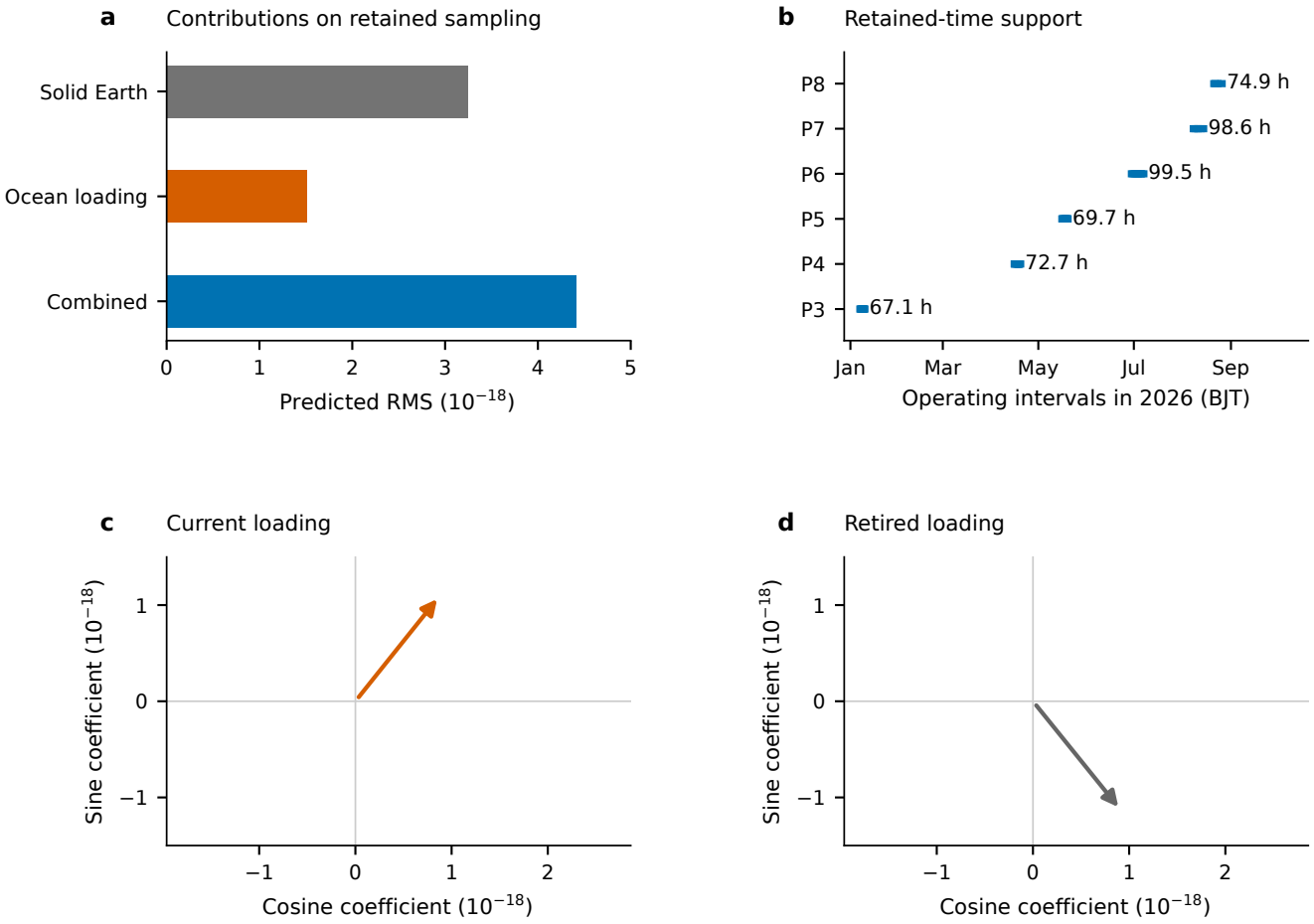
Extended Data Table 6 | Synthetic conditional validation on the frozen sampling. Each scenario has 512 realizations; error units are dimensionless template scale. D1 denotes the period-reference scalar lunar fit, D2 its period-daily alternative, and D3 the two-coefficient period-daily fit. D4 is the whole-template scalar, D5 the M2-only phase fit, and D6 M2 phase with period-daily terms. D1–D3 use seed 2026110501; D4–D6 use 2026110503. Errors are shared within, not across, these three-design batches. Noise codes are G, independent Gaussian; O, white plus OU; and T, Student-t plus MA. Max-SE means max(LS, delete-day, delete-period). Bias, SD and marginal-coverage columns concern the scalar or in-phase coefficient. Wild acceptance is scalar for D1, D2 and D4, and joint for D3, D5 and D6. MC SE is a binomial Monte Carlo standard error in percentage points; it is not experimental uncertainty. Noise scenarios were locally specified before these simulations, after the experimental analysis; they were not prospectively preregistered and do not certify the real noise process.

Design	Noise	Bias	SD	Mean max-SE	1.96 max-SE coverage	Wild acceptance	MC SE
D1	G	0.0031	0.1141	0.1331	96.68%	97.46%	0.70
D2	G	0.0112	0.1795	0.2328	97.07%	97.27%	0.72
D3	G	0.0111	0.1784	0.2354	97.27%	96.48%	0.81
D1	O	-0.0089	0.2400	0.3004	98.44%	96.29%	0.84
D2	O	0.0126	0.4051	0.6001	99.22%	95.70%	0.90
D3	O	0.0155	0.4068	0.6090	99.41%	95.70%	0.90
D1	T	-0.0043	0.1017	0.1212	97.66%	95.70%	0.90
D2	T	-0.0024	0.1595	0.2076	98.05%	96.48%	0.81
D3	T	-0.0025	0.1596	0.2093	98.24%	95.12%	0.95
D4	G	-0.0002	0.0882	0.0963	95.51%	96.09%	0.86
D5	G	-0.0028	0.1229	0.1433	96.48%	94.92%	0.97
D6	G	-0.0025	0.1923	0.2666	98.44%	96.09%	0.86
D4	O	-0.0066	0.2114	0.2429	95.12%	96.68%	0.79
D5	O	-0.0031	0.2536	0.3230	97.85%	96.68%	0.79
D6	O	-0.0038	0.4239	0.7350	99.80%	96.48%	0.81
D4	T	-0.0004	0.0703	0.0898	99.41%	96.68%	0.79
D5	T	0.0004	0.1036	0.1293	97.85%	96.09%	0.86
D6	T	0.0029	0.1650	0.2365	98.83%	96.88%	0.77

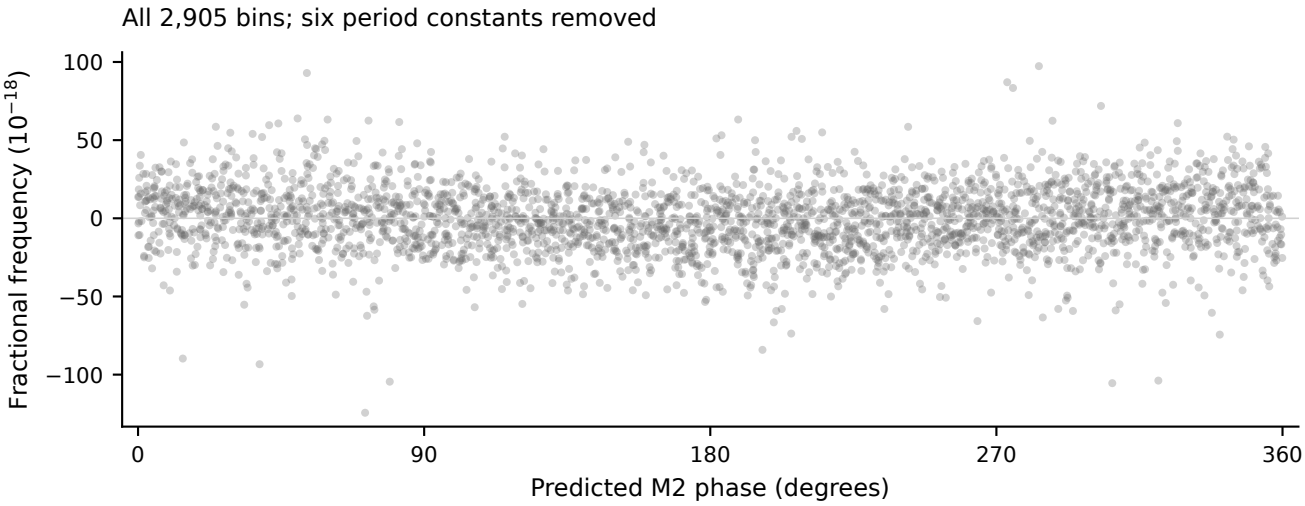
Grouped-lunar contour calibration | Conditional coverage of the earlier grouped-lunar phase contours. Each entry is the joint true-point coverage in 512 synthetic realizations, followed by its Monte Carlo standard error in percentage points. These use the actual delete-day covariance and unchanged $2F_{2,34}(0.95)$ contour threshold, seed 2026110504. G denotes independent Gaussian noise, O white plus OU noise, and T standardized Student-t plus MA noise as defined in Methods. These are not period-wild acceptance rates or real-noise coverage guarantees.

Noise	Period reference (%)	Period daily (%)	Ten-day daily (%)
G	94.53 (1.00)	95.70 (0.90)	96.29 (0.84)
O	97.07 (0.75)	98.05 (0.61)	99.41 (0.34)
T	95.51 (0.92)	98.05 (0.61)	97.07 (0.75)

These last three rows concern the grouped-lunar estimator, not the separate-M2 contours in the present Fig. 2b.

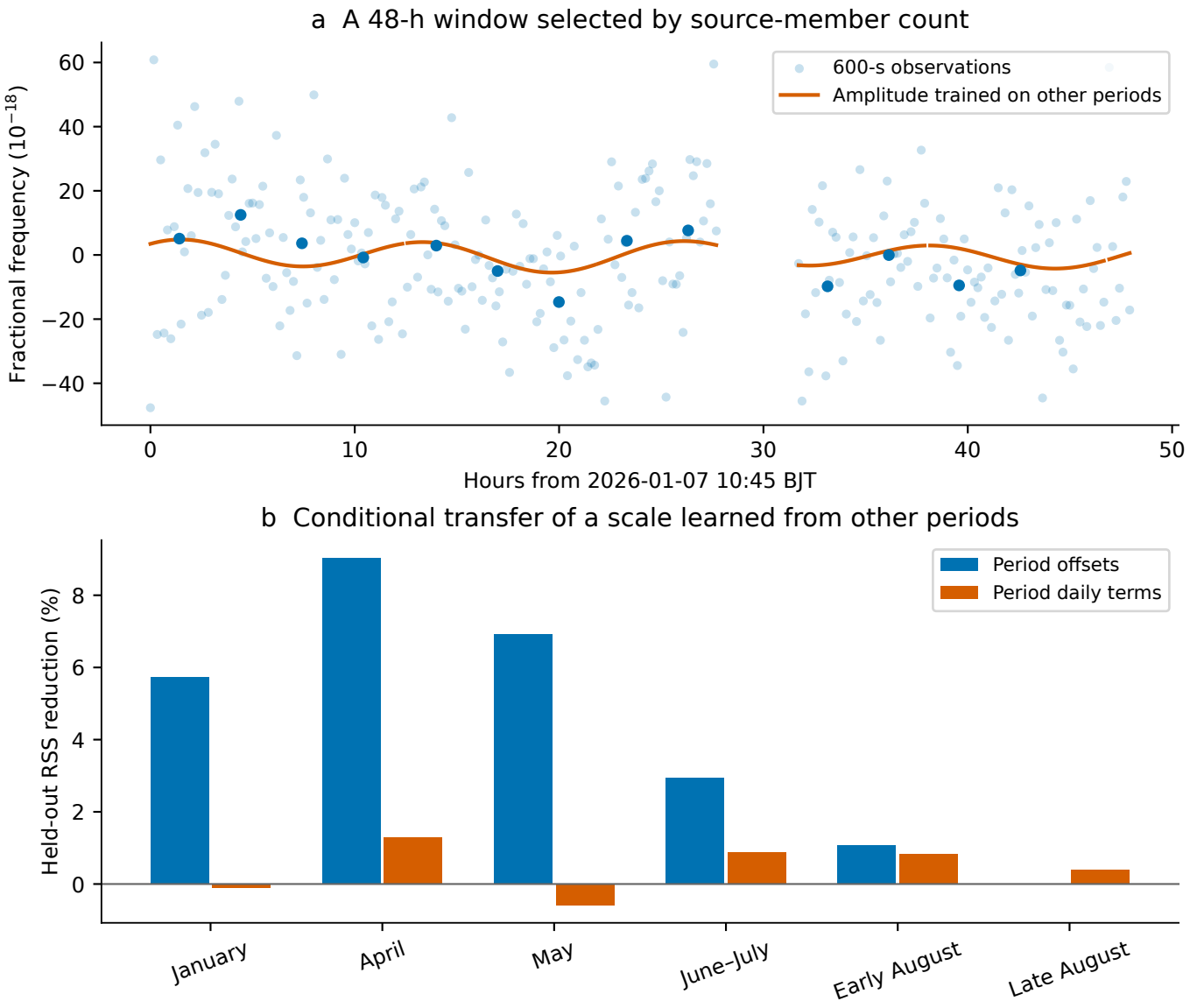


Extended Data Fig. 1 | Prediction, retained sampling and loading-synthesis provenance. a, Globally demeaned RMS of solid-Earth, loading and combined predictions on the same 2,905 averages. Components are correlated, so their RMS values are not additive. **b,** Effective-interval spans in the six periods. Labels count nominal retained members divided by 3,600; they are not verified uninterrupted uptime or statistical effective sample size. **c,d,** Current and retired loading M2 vectors, respectively, projected onto the same astronomical cosine/sine basis, in 10^{-18} fractional-frequency units before nodal modulation. Both axes use identical scales and equal physical units. Their coordinates are (0.8843, 1.1063) and (0.9227, −1.1437); the sine component changes sign. These are model vectors, not independent observations or confidence regions. The old synthesis pairs D^* with an unreversed $\exp(i\theta)$ and fixes the nodal factor to one. For the combined solid-Earth-plus-loading M2 templates, the retired/current amplitude ratio is 0.815 and the lag difference is −27.74°. Exclusion rests on that synthesis inconsistency: the six-period unit-point and free-amplitude phase tests give P=0.0625 and 0.09375, respectively, and do not reject the old phase at 5%. The local-time phase-coverage display appears in Fig. 1c.



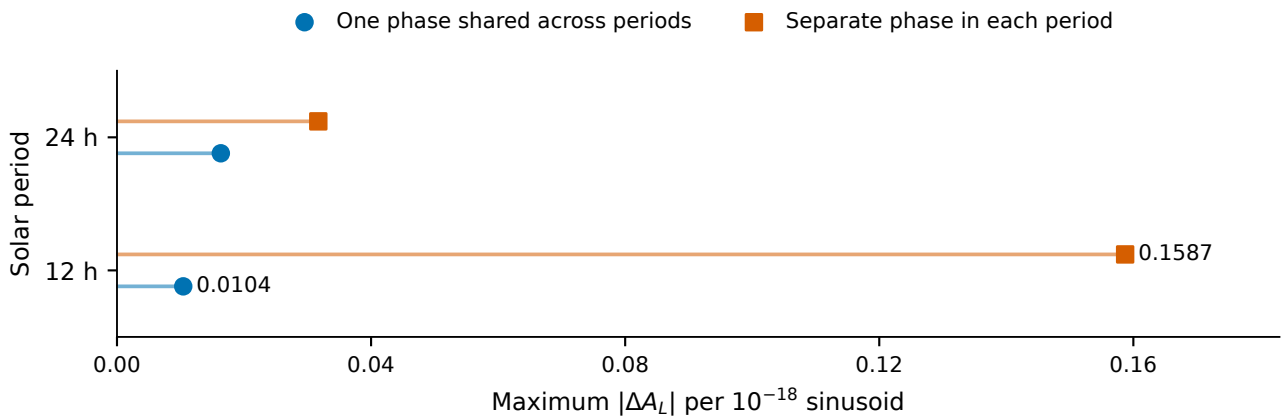
740

741 **Extended Data Fig. 2 | Complete scatter underlying the phase means.** All 2,905 averages after
742 subtracting the six period constants from the joint separate-M2/N2 model; no additional exclusions
743 or periodic-term subtraction. The full observed vertical range is retained. These are the same points
744 summarized in Fig. 2a, not an additional sample. No fit-spliced long-term stability is inferred.



745

746 **Extended Data Fig. 3 | Display window and conditional amplitude transfer. a,** The previously
747 selected 48-h window with the largest number of retained members, displayed with its period constant
748 removed and boundary-respecting averaging. **b,** Leave-one-period-out RSS changes for whole-template
749 reference and period-daily sensitivity models. Training data overlap, and test-period levels are fitted
750 locally. Values and selection rule are retained unchanged; neither display nor RSS improvement is an
751 independent detection. Detailed transfer calculations appear in Methods and Extended Data Table 3.



Extended Data Fig. 4 | Solar-day phase-coherence sensitivity. Maximum absolute change in the grouped M2+N2 scale for a prescribed 12-h or 24-h sinusoid of amplitude 10^{-18}. Blue fixes one phase across all periods; orange permits a separate phase in each period. Values are deterministic operator gains, not measured environmental amplitudes or uncertainty bars. The separate-M2 gains are reported in Methods; these curves and bars retain their original grouped-lunar estimand.